

THE MEMORY JOGGER™

A Pocket Guide
of Tools for
Continuous Improvement





THE
MEMORY
JOGGERTM

A Pocket Guide
of Tools for
Continuous Improvement

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GOAL/QPC

is a nonprofit organization striving to help companies continuously improve their **Quality, Productivity, and Competitiveness.**

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This handbook is designed to help you and every person in your company to IMPROVE DAILY THE PROCEDURES, SYSTEMS, QUALITY, COST, AND YIELDS RELATED TO YOUR JOB. This continuous improvement process is the focus of today's QUALITY REVOLUTION.

In companies that are involved in this revolution, this continuous improvement process has two components:

- 1) Philosophy
- 2) Problem-Solving/Graphical Techniques

1. PHILOSOPHY

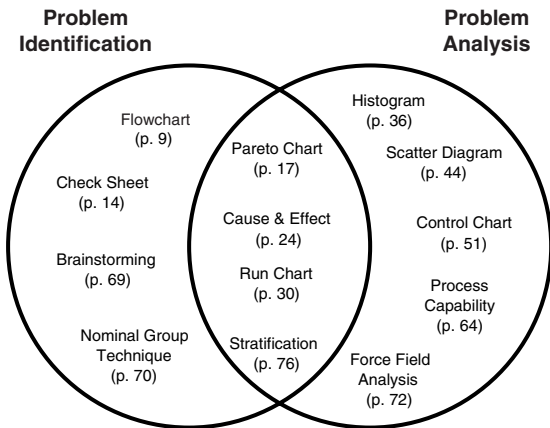
There are common points in the operating philosophies of these companies. They are as follows:

- Improving quality by removing the causes of problems in the system **inevitably** leads to improved productivity.
- The person doing the job is most knowledgeable about that job.
- People want to be involved and do their jobs well.
- Every person wants to feel like a valued contributor.
- More can be accomplished by working together to improve the system than having individual contributors working around the system.
- A structured problem-solving process using graphical techniques produces better solutions than an unstructured process.
- Graphical problem-solving techniques let you know where you are, where the variations lie, the relative importance of problems to be solved, and whether the changes made have had the desired impact.

- The adversarial relationship between labor and management is counterproductive and outmoded.
- Every organization has undiscovered “gems” waiting to be developed.

2. PROBLEM-SOLVING/GRAPHICAL TECHNIQUES

The rest of this handbook consists of practical descriptions, instructions, and examples of the following techniques:



Notice that charts can be used for different purposes in various stages of the problem-solving process. For example, the tools included in the intersecting portion of this **VENN DIAGRAM** can be used in both the Problem Identification and Problem Analysis phase of problem solving.

Task

1. Decide which problem will be addressed first (or next).
2. Arrive at a statement that describes the problem in terms of what it is specifically, where it occurs, when it happens, and its extent.
3. Develop a complete picture of all the possible causes of the problem.
4. Agree on the basic cause(s) of the problem.
5. Develop an effective and implementable solution and action plan.
6. Implement the solution and establish needed monitoring procedures and charts.

SELECTION GUIDE

Techniques

- Flow Chart (p. 9)
- Check Sheet (p. 14)
- Pareto Chart (p. 17)
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- Nominal Group Technique (p. 70)
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- Stratification (p. 76)

HOW TO USE *THE MEMORY JOGGER*™

The Memory Jogger™ is designed as a convenient and quick reference guide on the job. It is therefore organized around symbols and color highlights that are eye-catching and easy to remember. The following is a legend explaining each symbol and its use.

Making Ready An important step in a problem-solving process is selecting the right tool for the situation. Each section on a new tool will begin with a boxed description that describes when it should be used. Always check this feature first to ensure that the tool meets your needs.



Cruising



Turn to this portion of each section to find construction guidelines. This is the action phase that provides you with step-by-step instructions and helpful formulas. Turn to this feature in each section when you have basic how-to questions.

Finishing The Course



This portion of each section shows each tool in its final form. There are examples from **Manufacturing, Administration/Service, and Daily Life** to display the widespread applications of each tool. Refer to this portion when you need to see the proper form of the finished charting technique.

Caution



The boxed portion at the end of each section describes helpful construction and interpretation tips for that charting technique. Be sure to use this section to avoid making some of the most common mistakes when constructing and analyzing these tools.

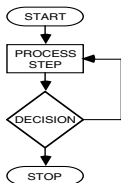


Flowchart: When you need to identify the actual ideal paths that any product or service follows in order to identify deviations.



FLOWCHART

A Flowchart is a pictorial representation showing all of the steps of a process. Flowcharts provide excellent documentation of a program and can be a useful tool for examining how various steps in a process are related to each other. Flowcharting uses easily recognizable symbols to represent the type of processing performed.



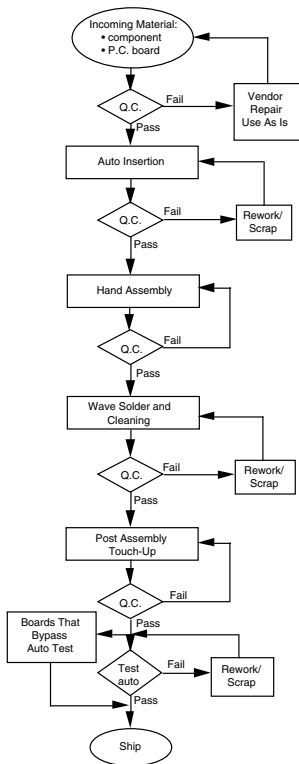
By studying these charts you can often uncover loop-holes which are potential sources of trouble. Flowcharts can be applied to anything from the travels of an invoice or the flow of materials, to the steps in making a sale or servicing a product.

The Flowchart is most widely used in problem identification in a process called IMAGINEERING. The people with **important** knowledge about the process meet to:

- 1.) Draw a Flowchart of what steps the process **actually** follows.
- 2.) Draw a Flowchart of what steps the process **should** follow if everything worked right.
- 3.) Compare the two charts to find where they are different because this is where the problems arise.

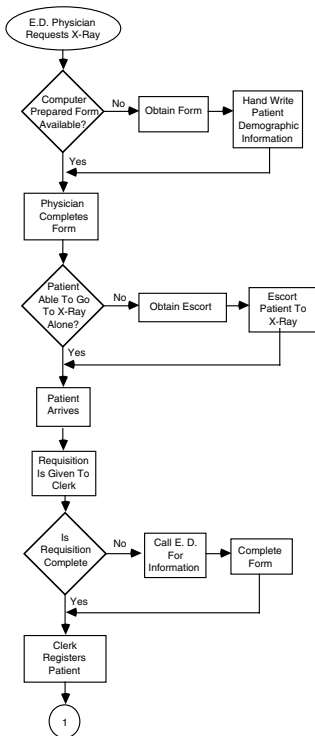
Flowchart—Mfg. Example

Board Flow



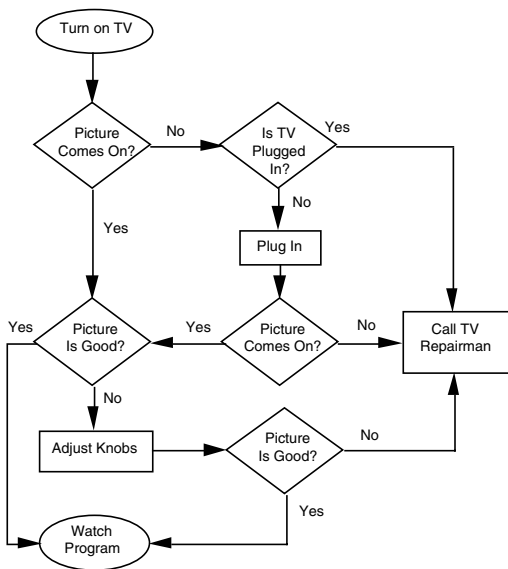
Flowchart—Admin./Service Example

Emergency Department/Radiology Process



Flowchart—Daily Example

Turning On A Television





Construction/Interpretation Tips Flowchart

- Define the boundaries of the process clearly.
- Use the simplest symbols possible.
- Make sure every feedback loop has an escape.
- There is usually only one output arrow out of a *process* box. Otherwise, it may require a *decision* diamond.



Check Sheet: When you need to gather data based on sample observations in order to begin to detect patterns. This is the logical point to start in most problem-solving cycles.



CHECK SHEET

Check Sheets are simply an easy to understand form used to answer the question, “How often are certain events happening?” It starts the process of translating “opinions” into “facts.” Constructing a Check Sheet involves the following steps:

- 1.) Agree on the event being observed. Everyone has to be looking for the same thing.
- 2.) Decide on a time period to collect data. This could range from hours to weeks.
- 3.) Design a form that is clear and easy to use, with columns clearly labeled and adequate space for entering data.
- 4.) Collect the data consistently and honestly. Make sure there is time allowed for this data-gathering task.

Problem	Month			
	1	2	3	Total
A	II	II	I	5
B	I	I	I	3
C	IIII	II	IIII	12
Total	8	5	7	20

Check Sheet—Mfg. Example Bearing Defects



Defect	May				Total
	6	7	8	9	
Wrong Size	I				26
Wrong Shape	I				9
Wrong Dept.		I	I	I	8
Wrong Weight					52
Wrong Smoothness			I	I	7
Total	29	22	25	26	102

Check Sheet—Admin./Service Ex. Typing Mistakes In Department A

Mistakes	March			Total
	1	2	3	
Centering				8
Spelling		I		23
Punctuation				40
Missed Paragraph		I	I	4
Wrong Numbers				10
Wrong Page #'s	I	I		4
Tables				13
Total	34	35	33	102

Check Sheet—Daily Example Reasons For Disagreements

Reason	Day					Total
	M	T	W	T	F	
Money						20
Sex						10
Children						19
Total	12	6	10	8	13	49



Construction/Interpretation Tips Check Sheet

- Make sure that observations/samples are as representative as possible.
- Make sure the sampling process is efficient so that people have time to do it.
- In doing a Check Sheet, you may not be aware that the population (universe) being sampled is non-homogeneous (not from the same machine, person, etc.). The population must be homogeneous. If not, it must first be stratified (grouped), with each grouping sampled individually.



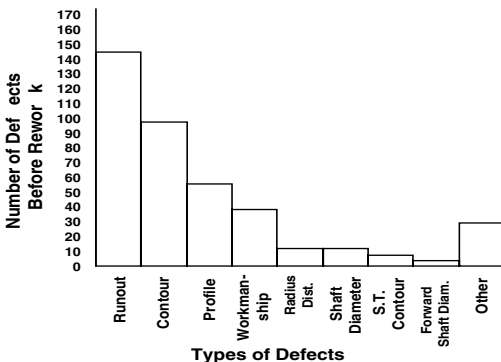
Pareto Chart: When you need to display the relative importance of all of the problems or conditions in order to: choose the starting point for problem solving, monitor success, or identify the basic cause of a problem.



PARETO CHART

A Pareto Chart is a special form of vertical bar graph which helps us to determine which problems to solve in what order. Doing a Pareto Chart based upon either Check Sheets or other forms of data collection helps us direct our attention and efforts to the truly important problems. We will generally gain more by working on the tallest bar than tackling the smaller bars.

Pareto Chart Defects Found At In-Process Inspection



STEPS IN CONSTRUCTING A PARETO CHART

- 1.) Select the problems that are to be compared and rank-ordered by:
 - a.) **Brainstorming**, e.g., "What are our major quality problems in Department A?"
 - b.) **Using Existing Data**, e.g., "Let's take a look at Department A's quality reports over the last month to find the major problem areas."
- 2.) Select the standard for comparison unit of measurement, e.g., annual cost, frequency, etc.
- 3.) Select time period to be studied, e.g., 8 hours, 5 days, 4 weeks.
- 4.) Gather necessary data on the site of each category, e.g., "Defect A occurred X times in the last 6 months" or "Defect B cost X dollars in the last 6 months."
- 5.) Compare the frequency or cost of each category relative to all other categories, e.g., "Defect A happened 75 times; Defect B happened 107 times; Defect C happened 35 times," or "Defect A cost \$750 annually; Defect B cost \$535 annually."
- 6.) List the categories from left to right on the horizontal axis in their order of decreasing frequency or cost. The categories containing the fewest items can be combined into an "other" category, which is placed on the extreme right as the last bar.
- 7.) Above each classification or category, draw a rectangle whose height represents the frequency or cost in that classification.*

* Additional Features of Pareto Charts:

- Often the “raw data” is recorded on the left vertical axis with a percentage scale on the right vertical axis. Make sure that the two axes are drawn to scale, e.g., 100% is opposite the total frequency or cost; 50% is opposite the halfway point in the raw data.
- From the top of the tallest bar and moving upward from left to right, a line can be added that shows the cumulative frequency of the categories. This answers such questions as, “How much of the total is accounted for by the first three categories?”

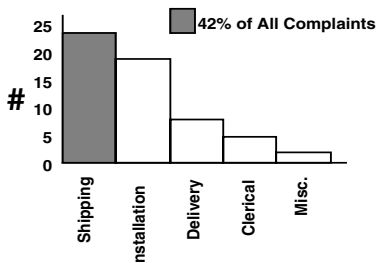
DIFFERENT USES OF A PARETO CHART



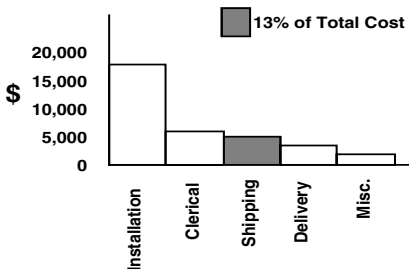
- 1.) To identify the most important problems through the use of different measurement scales, e.g., frequency, cost.

Lesson: The most frequent problems are not always the most costly.

Field Service Customer Complaints:



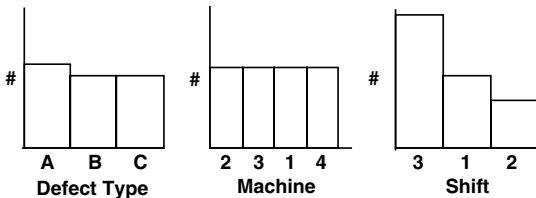
Cost To Rectify Field Service Complaints:



- 2.) To analyze different groupings of data, e.g., by product, by machine, by shift.

Lesson: If clear differences don't emerge, regroup the data. Use your imagination.

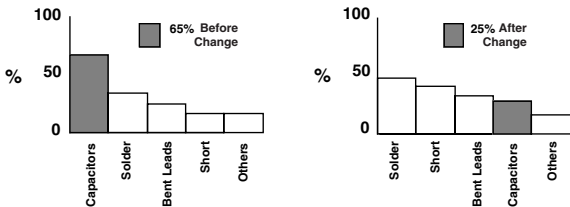
Pareto Analysis Of The Number Of Defects



- 3.) To measure the impact of changes made in a process, e.g., before and after comparisons.

Lesson: You don't know how much better you are if you don't know where you were before the change.

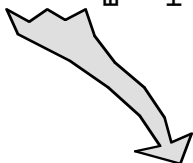
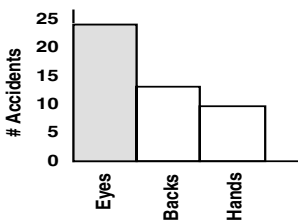
Component Defects (Station #2)



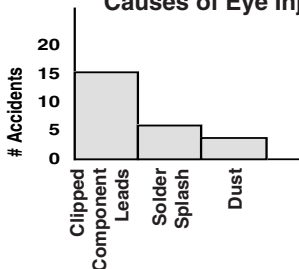
- 4.) To break down broad causes into more and more specific parts.

Lesson: Cure the cause, not the symptom.

Types of Injury



Causes of Eye Injury





Construction/Interpretation Tips Pareto Chart

- Use common sense—two key customer complaints may deserve more attention than 100 other complaints, depending on who the customer is and what the complaint is.
- Mark chart clearly to show standard of measurement (\$, %, or #).

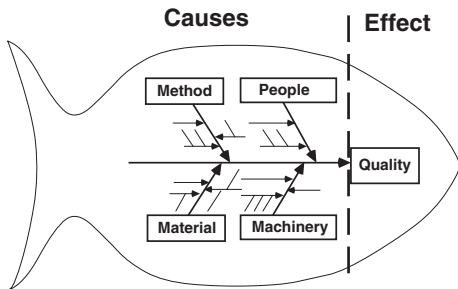


Cause & Effect Diagram: When you need to identify and explore and display the possible causes of a specific problem or condition.



CAUSE & EFFECT DIAGRAM

The Cause & Effect Diagram was developed to represent the relationship between some “effect” and all the possible “causes” influencing it. The effect or problem is stated on the right side of the chart and the major influences or “causes” are listed to the left.



(Fishbone Diagram)

Cause & Effect Diagrams are drawn to clearly illustrate the various causes affecting a process by sorting out and relating the causes. For every effect there are likely to be several major categories of causes. The major causes might be summarized under four categories, People, Machines, Methods, and Materials. In administrative areas it may be more helpful to use the 4P's: Policies, Procedures, People, and Plant. Remember that these categories are only suggestions. You may use any major category that emerges or helps people think creatively.

A well-detailed Cause & Effect Diagram will take on the shape of fishbones and hence the alternate name Fishbone Diagram. From this well-defined list of possible causes, the most likely are identified and selected for further analysis. When examining each cause look for things that have changed, deviations from the norm or patterns. Remember, look to cure the cause and not the symptoms of the problem. Push the causes back as much as is practically possible.

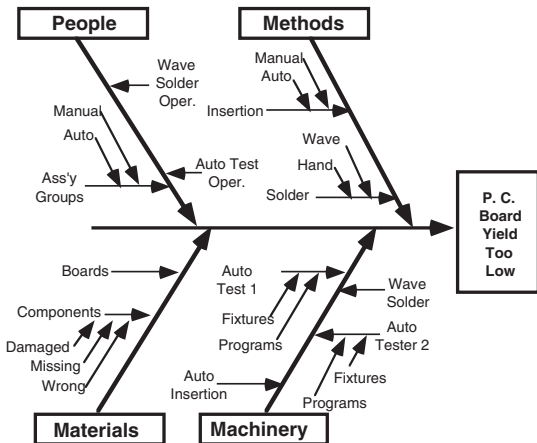
STEPS IN CONSTRUCTING A CAUSE & EFFECT DIAGRAM

- 1.) Generate the causes needed to build a Cause & Effect Diagram in one of two ways:
 - a.) Structured **Brainstorming** about possible causes without previous preparation.
 - b.) Ask members of the team to spend time between meetings using simple **Check Sheets** to track possible causes and to examine the production process steps closely.
- 2.) Construct the actual Cause & Effect Diagram by:
 - a.) Placing problem statement in box on the right.
 - b.) Drawing the traditional major cause category steps in the production process, or any causes that are helpful in organizing the most important factors.
 - c.) Placing the **Brainstormed** ideas in the appropriate major categories.
 - d.) For each cause ask, "Why does it happen?" and list responses as branches off the major causes.
- 3.) Interpretation

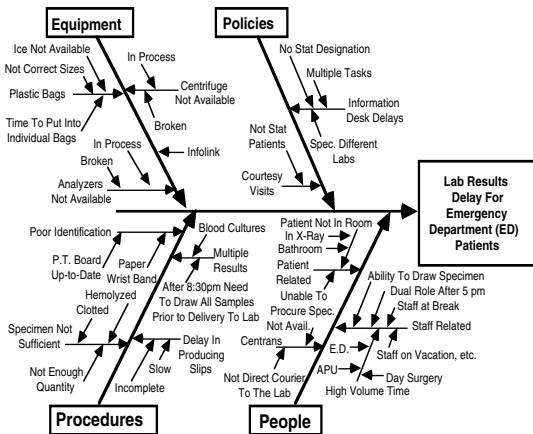
In order to find the most basic causes of the problem:

 - a.) Look for causes that appear repeatedly.
 - b.) Reach a team consensus.
 - c.) Gather data to determine the relative frequencies of the different causes.

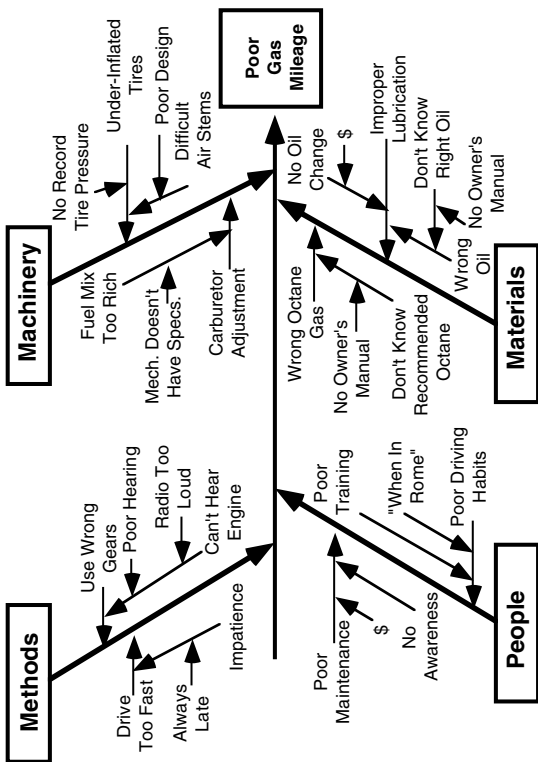
Cause & Effect Diagram— Manufacturing Example



Cause & Effect Diagram— Admin./Service Example



Cause & Effect Diagram—Daily Example





Construction/Interpretation Tips Cause & Effect Diagram

- Try not to go far beyond the area of control of the group in order to minimize frustration.
- If ideas are slow in coming use the major cause categories as catalysts, e.g., “What in *materials* is causing...?”
- Use as few words as possible.
- Make sure everyone agrees completely on the problem statement.
- The most widely used type of Cause & Effect Diagram is the **Dispersion Analysis** type which is the type shown in *The Memory Jogger™*. It is constructed by placing individual causes within each “major” cause category and asking the following question of each item, “Why does this cause (dispersion) happen?” Other common types of Cause & Effect Diagrams are as follows:
 - a. **Process Classification Cause & Effect Diagrams** sequentially list all the steps in a process. The same cause category arrows as in the Dispersion Analysis type branch off the line between each process step. The same questions are then applied to each cause category as in the Dispersion Analysis type diagram.
 - b. **Cause Enumeration Cause & Effect Diagrams** are almost identical to the Dispersion Analysis type. The only real difference rests in the fact that Cause Enumeration first organizes all the possible causes in list form and then places them in the major cause categories.



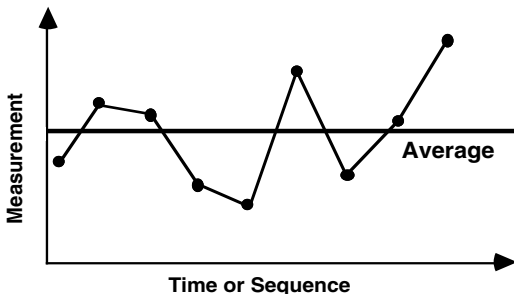
Run Chart: When you need to do the simplest possible display of trends within observation points over a specified time period.



RUN CHART

Run Charts are employed to visually represent data. They are used to monitor a process to see whether or not the long range average is changing.

Run Charts are the simplest tool to construct and use. Points are plotted on the graph in the order in which they become available. It is common to graph the results of a process such as machine downtime, yield, scrap, typographical errors, or productivity as they vary over time.



A danger in using a Run Chart is the tendency to see every variation in data as being important. The Run Chart, like the other charting techniques, should be **used to focus attention on truly vital changes in the process.**

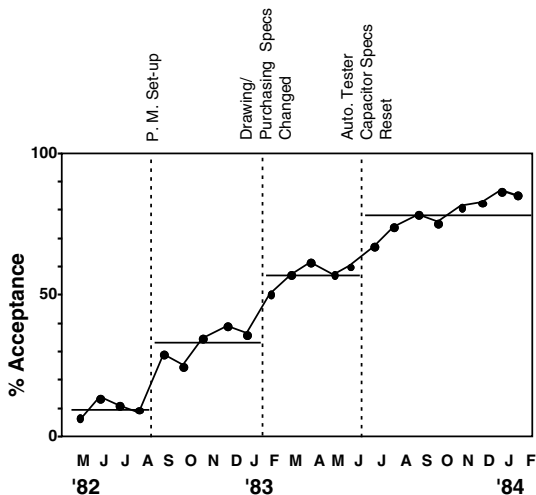
One of the most valuable uses of Run Charts is to identify meaningful trends or shifts in the average. For example when monitoring any process, it is expected that we should find an equal number of points falling above and below the average. Therefore, when nine points “run” on one side of the average it indicates a statistically unusual event and that the average has changed. Such changes should always be investigated. If the shift is favorable, it should be made a permanent part of the system. If it is unfavorable, it should be eliminated.

An alternate type of pattern that can occur is a trend of six or more points steadily increasing or decreasing with no reversals. Neither pattern would be expected to happen based on random chance. Thus, this would likely indicate an important change and the need to investigate.

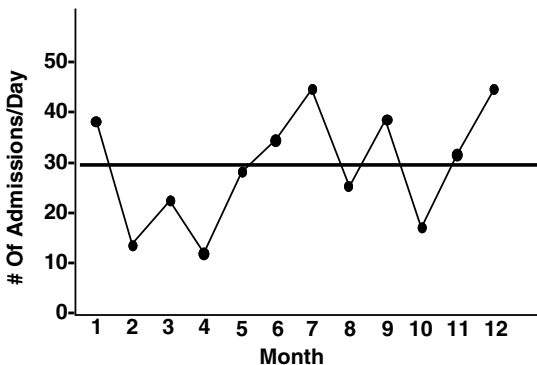
Run Chart—Mfg. Example

% Accept On First Test

Printed Circuit Boards

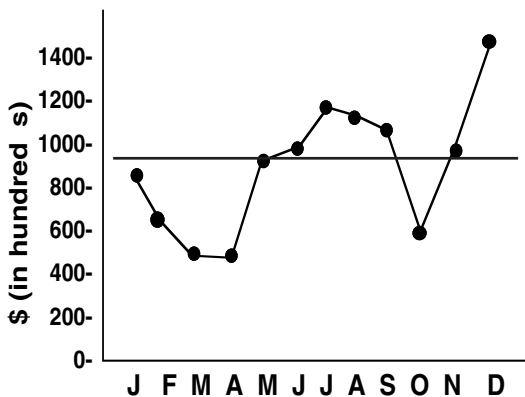


Run Chart—Admin./Service Example Emergency Room Admissions



Run Chart—Daily Example

Family Expenditure/Month





Construction/Interpretation Tips Run Chart

- The *y axis* is the vertical side of the graph.
- The *x axis* is the horizontal side of the graph.
- A marked point indicates the measurement or quantity observed or sampled at one point in time.
- Data points should be connected for easy use and interpretation.
- The time period covered and unit of measurement used must be clearly marked.
- Collected data must be kept in the order that it was gathered. Since it is tracking a characteristic over time, the sequence of data points is critical.

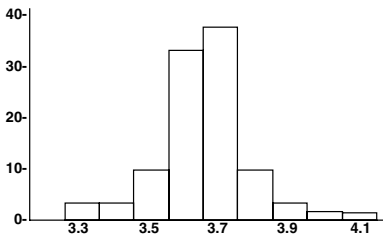


Histogram: When you need to discover and display the distribution of data by bar graphing the number of units in each category.



HISTOGRAM

As we have already seen with the **Pareto Chart**, it is very helpful to display in bar graph form the frequency with which certain events occur (frequency distribution). The **Pareto Chart**, however, only deals with characteristics of a product or service, e.g., type of defect, problem, safety hazards (attribute data). A Histogram takes measurement data, e.g., temperature, dimensions, and displays its distribution. This is critical since we know that all repeated events will produce results that vary over time. A Histogram reveals the amount of variation that any process has within it. A typical Histogram would look like this:

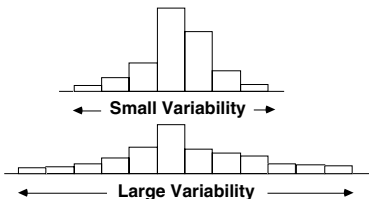


The Histogram pictured here shows the greatest number of units are at the center measurement with roughly an equal number of units falling on either side of that point. Many repeated samples of data under statistical control follow this pattern. Other data displays patterns

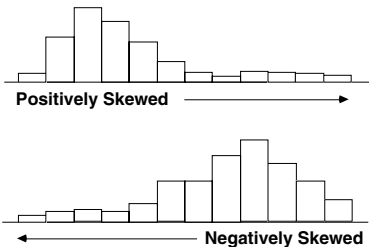
with the data “piled up” at points away from the center. Such a distribution is referred to as being “skewed.” The important thing to remember is that you are looking for surprises such as distributions that should be naturally “normal” but are not. The same is true for predictably skewed distributions. In addition to the shape of the distribution you are also looking for:

- a.) Whether the “spread” of the curve falls within specifications. If not, how much falls outside of specifications. (VARIABILITY)
- b.) Whether the curve is centered at the right place. Are most items on the “high or low side?” (SKEWNESS)

Illustrations Of Variability



Illustrations Of Skewness



STEPS IN CONSTRUCTING A HISTOGRAM

There will be more detailed instructions for constructing Histograms than most of the other tools. This is being stressed because of the confusion that seems to be created when deciding on the number of classes (bars), the class boundaries, etc.

Start with an unorganized set of numbers such as the following:

9.9	9.3	10.2	9.4	10.1	9.6	9.9	10.1	9.8
9.8	9.8	10.1	9.9	9.7	9.8	9.9	10.0	9.6
9.7	9.4	9.6	10.0	9.8	9.9	10.1	10.4	10.0
10.2	10.1	9.8	10.1	10.3	10.0	10.2	9.8	10.7
9.9	10.7	9.3	10.3	9.9	9.8	10.3	9.5	9.9
9.3	10.2	9.2	9.9	9.7	9.9	9.8	9.5	9.4
9.0	9.5	9.7	9.7	9.8	9.8	9.3	9.6	9.7
10.0	9.7	9.4	9.8	9.4	9.6	10.0	10.3	9.8
9.5	9.7	10.6	9.5	10.1	10.0	9.8	10.1	9.6
9.6	9.4	10.1	9.5	10.1	10.2	9.8	9.5	9.3
10.3	9.6	9.7	9.7	10.1	9.8	9.7	10.0	10.0
9.5	9.5	9.8	9.9	9.2	10.0	10.0	9.7	9.7
9.9	10.4	9.3	9.6	10.2	9.7	9.7	9.7	10.7
9.9	10.2	9.8	9.3	9.6	9.5	9.6	10.7	

These numbers refer to the thickness of a certain key material in a process.

STEP 1: Count the number of data points in the set data.

For our example above there are 125 data points ($n=125$).

STEP 2: Determine the range, R, for the entire data set.

The range is the smallest value in the set of data subtracted from the largest value. In our case, the range is equal to 10.7 minus 9.0. Thus, the range equals 1.7.

STEP 3: Divide the data set¹ into a certain number of classes, referred to as K. The table below provides an approximate guideline for dividing your set of data into a reasonable number of classes. For our example, 125 data points would be broken down into 7-12 classes. We will use K=10 classes.

<u>Number Of Data Points</u>	<u>Number Of Classes (K)</u>
Under 50	5-7
50-100	6-10
100-250	7-12
Over 250	10-20

STEP 4: Determine the class width, H. A convenient formula is as follows:

$$H = \frac{R}{K} = \frac{1.7}{10} = .17$$

In this case, as in most, it helps to round off H but carry the number to one more decimal point than in the original data set. For our purposes, 0.20 would appear appropriate.

¹ In previous printings this was referred to as "range value". It has been revised for greater clarity.

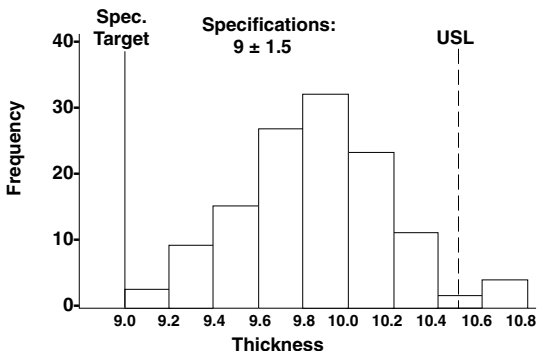
STEP 5: Determine the class boundary, or end points.

For simple determination of the class boundaries, take the smallest individual measurement in the data set. Use this number or round to the next appropriate lower number. This will be the lower end point for our first class boundary. In our example this would be 9.00. Now take this number and add the class width to it, $9.00 + 0.20 = 9.20$. Thus, the next lower class boundary would begin at 9.20. The first class would be 9.00 and everything up to, **but not including** 9.20, 9.00 through 9.19! The second class would begin at 9.20 and be everything up to, but not including 9.40! This makes each class mutually exclusive, that is each of our data set points will fit into one, **and only one**, class. Finally, consecutively add the class width, 0.20, to the lowest class boundary until the correct number of classes, approximately 10 and containing the range of all our numbers, is obtained.

STEP 6: Construct a frequency table based on the values computed above (i.e., number of classes, class width, class boundary). The frequency table is actually a Histogram in a tabular form. A frequency table based on the thickness data is shown below:

Class #	Class Boundaries	Mid-point	Frequency	Total
1	9.00-9.19	9.1		1
2	9.20-9.39	9.3		9
3	9.40-9.59	9.5		16
4	9.60-9.79	9.7		27
5	9.80-9.99	9.9		31
6	10.00-10.19	10.1		22
7	10.20-10.39	10.3		12
8	10.40-10.59	10.5		2
9	10.60-10.79	10.7		5
10	10.80-10.99	10.9		0

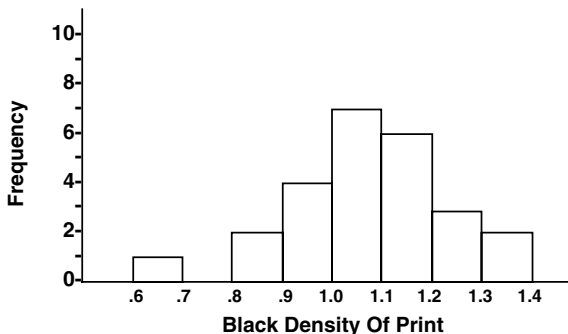
STEP 7: Construct the Histogram based on the frequency table. A Histogram is a graphical picture of a frequency table. It provides us with a quick picture of the distribution for the measured characteristic. A Histogram for our example is shown below:



As pointed out earlier, the Histogram is an important diagnostic tool because it gives a “bird's-eye view” of the variation in a data set. In our case, the data appears to have a central tendency around 9.8 to 9.99. It also appears the data creates close to a normal curve. The specification for the thickness characteristic is 7.5 to 10.5, with a target of 9. Thus, we can see that our Histogram indicates the process is running high and that defective material is being made.

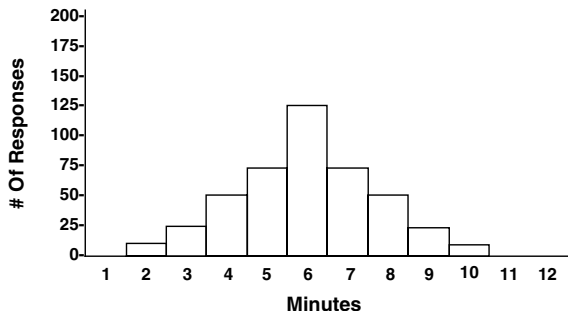
Histogram—Mfg. Example

Print Density



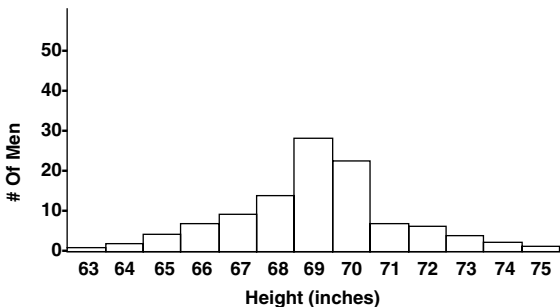
Histogram—Admin./Service Example

Average Response Time To Patient Rings (1st Shift)



Histogram—Daily Example

Height of 100 Men



Construction/Interpretation Tips

Histogram

- The number of classes (bars in the graph) determines how much of a pattern will be visible.
- Some processes are naturally skewed; don't expect every distribution to follow a bell-shaped curve.
- Get suspicious of the accuracy of the data if the classes suddenly stop at one point (such as a specification limit) without some previous decline in number.
- Always look for *twin peaks* indicating that the data is coming from two or more different sources, e.g., shifts, machines.



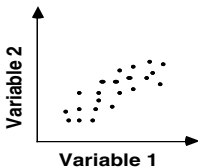
Scatter Diagram: When you need to display what happens to one variable when another variable changes in order to test a theory that the two variables are related.



SCATTER DIAGRAM

A Scatter Diagram is used to study the possible relationship between one variable and another. The Scatter Diagram is used to test for possible **cause** and **effect** relationships. It cannot prove that one variable **causes** the other, but it does make it clear whether a relationship exists and the strength of that relationship.

A Scatter Diagram has a horizontal axis (x-axis), to represent the measurement values of one variable, and a vertical axis (y-axis), to represent the measurements of the second variable. A “typical” Scatter Diagram may look like this:

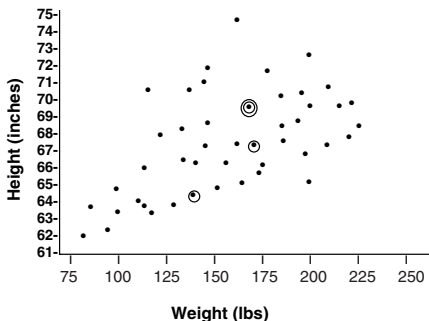


Notice how the plotted points form a clustered pattern. The direction and “tightness” of the cluster give you a clue as to the strength of the relationship between variable 1 and variable 2. The more that this cluster resembles a straight line, the stronger the correlation between the variables. This makes sense since a straight line would mean that every time one variable would change the other would change by the same amount.

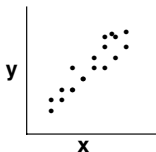
- 1.) Collect 50 to 100 paired samples of data that you think may be related and construct a data sheet as follows:

Person	Weight	Height
1	160 lbs.	70 inches
2	180 "	61 "
3	220 "	75 "
.	.	.
.	.	.
50	105 "	61 "

- 2.) Draw the horizontal and vertical axes of the diagram. The values should get higher as you move up and to the right on each axis. The variable that's being investigated as the possible "cause" is usually on the horizontal and the "effect" variable is usually on the vertical.
- 3.) Plot the data on the diagram. If you find the values being repeated, circle that point as many times as appropriate. The resulting diagram may look like this:



The following are the various patterns and meanings that Scatter Diagrams can have:

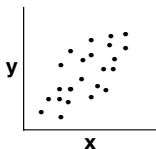


1. Positive correlation

1. An increase in y may depend on an increase in x. If x is controlled, we might have a good chance of controlling y.

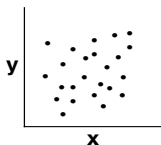
Example:

- training vs. performance



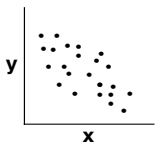
2. Possible positive correlation

2. If x is increased, y may increase somewhat, but y seems to have causes other than x.



3. No correlation

3. There may be no correlation. Y may be dependent on another variable.

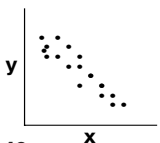


4. Possible negative correlation

4. An increase in x may cause a tendency for a decrease in y.

Examples:

- quality vs. customer complaints
- training vs. rejects

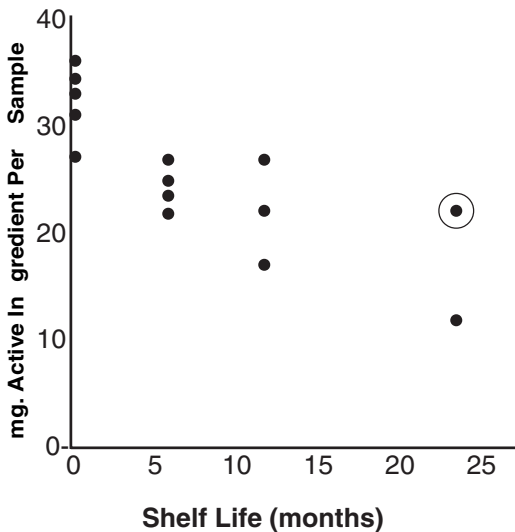


5. Negative correlation

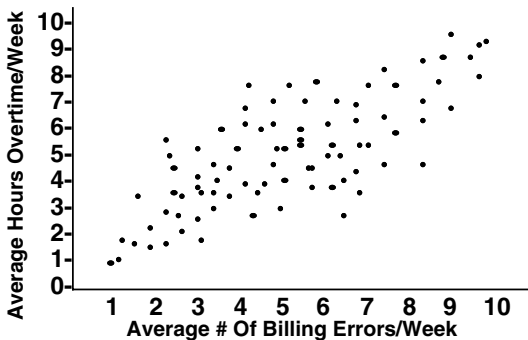
5. An increase in x may cause a decrease in y. Therefore, as with item 1 above, x may be controlled instead of y.

Scatter Diagram—Mfg. Example

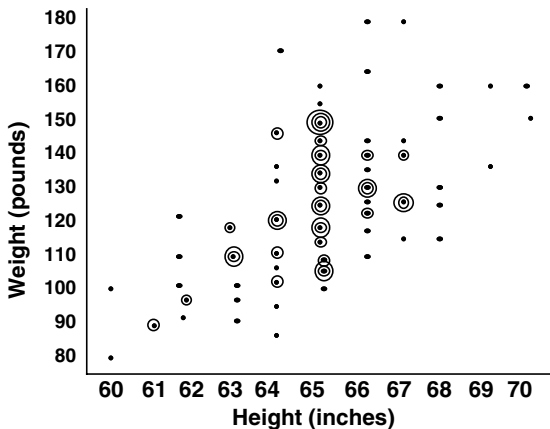
Active Ingredient Stability



Scatter Diagram—Admin./Service Ex. Overtime/# Of Billing Errors



Scatter Diagram—Daily Example





Construction/Interpretation Tips Scatter Diagram

- A negative relationship (as y increases, x decreases) is as important as a positive relationship (as x increases, y increases).
- You can only say that x and y are related and not that one *causes* the other.
- The examples in this section were based on straight line correlations: $y=a+bx$. However, this is not the only form of relationship that can be routinely encountered: $y=e^x$, $y=x^2$, and $y^2=x$ are just a few of the many types that can occur.
- There are statistical tests available to test the exact degree of correlation but are beyond the scope of this book.

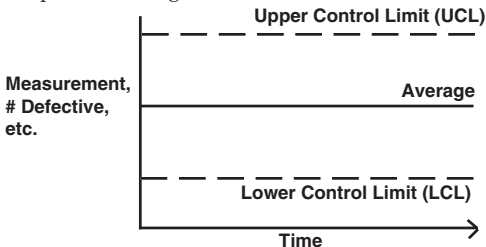


Control Chart: When you need to discover how much variability in a process is due to random variation and how much is due to unique events/individual actions in order to determine whether a process is in statistical control.



CONTROL CHART

A Control Chart is simply a run chart with **statistically** determined upper (Upper Control Limit) and possibly lower (Lower Control Limit) lines drawn on either side of the process average.

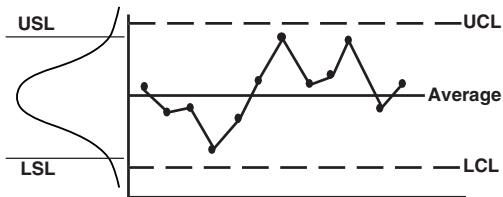


These limits are calculated by running a process untouched (i.e., according to standard procedures but without any extra "tweaking" adjustments), taking samples, and plugging the sample averages into the appropriate formula. You can now plot the sample averages onto a chart to determine whether any of the points fall between or outside of the limits or form unlikely patterns. If either of these happen, the process is said to be "out of control."

The fluctuation of the points within the limits results from variation built into the process. This results from **common causes** within the **system** (e.g., design, choice of

machine, preventive maintenance), and can only be affected by changing that system. However, points outside of the limits come from a **special cause** (e.g., people errors, unplanned events, freak occurrences), that is not part of the way that the process normally operates, or from an unlikely combination of process steps. These special causes must be eliminated before the Control Chart can be used as a monitoring tool. Once this is done, the process would be “in control” and samples can be taken at regular intervals to make sure that the process doesn’t fundamentally change.

REMEMBER: “Control” doesn’t necessarily mean that the product or service will meet your needs. It only means that the process is **consistent** (may be consistently bad). For example:



In this case, the process is in control but it is not capable of meeting the specification. The curve to the left of the Control Chart shows that the specification limits are narrower than the Control Charted process. Either you improve the process or you change the specifications. Just remember that specifications are what you think you need and control limits are what the process can do consistently.

STEPS IN CONSTRUCTION OF CONTROL CHARTS AND CRITICAL FORMULAS

Variables Control Chart:

When samples are expressed in **quantitative** units of measurement, e.g., length, weight, time.

$\bar{X}$ —R Chart =

Plotting the Average & Range of Data Collected

Calculate the Average ($\bar{X}$) and Range (R) of each subgroup:

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \quad n = \# \text{ of samples}$$

$$R = X_{\max} - X_{\min}$$

Calculate the Average Range ($\bar{R}$) and the Process Average ($\bar{\bar{X}}$):

$$\bar{\bar{X}} = \frac{\bar{X}_1 + \bar{X}_2 + \dots + \bar{X}_k}{k} \quad k = \# \text{ of subgroups}$$

$$\bar{R} = \frac{R_1 + R_2 + \dots + R_k}{k} \quad (20\text{-}25 \text{ groups})$$

Calculate the Control Limits:

$$UCL_{\bar{X}} = \bar{\bar{X}} + A_2\bar{R}$$

$$LCL_{\bar{X}} = \bar{\bar{X}} - A_2\bar{R}$$

$$UCL_R = D_4\bar{R}$$

$$LCL_R = D_3\bar{R}$$

Table of Factors for $\bar{X}$ & R Charts

Number of observations in subgroup (n)	Factors for X Chart	Factors for R Chart	
	A_2	Lower D_3	Upper D_4
2	1.880	0	3.268
3	1.023	0	2.574
4	0.729	0	2.282
5	0.577	0	2.114
6	0.483	0	2.004
7	0.419	0.076	1.924
8	0.373	0.136	1.864
9	0.337	0.184	1.816
10	0.308	0.223	1.777

STEPS IN CONSTRUCTION OF CONTROL CHARTS AND CRITICAL FORMULAS

Attributes Control Chart:

When sample reflects **qualitative** characteristics, e.g., is/is not defective, go/no go.

The p Chart = Proportion Defective

$p = \frac{\text{number of rejects in subgroups}}{\text{number inspected in subgroup}}$

$\bar{p} = \frac{\text{total number rejects}}{\text{total number inspected}}$

$$UCL_p^* = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \qquad LCL_p^* = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$$

The np Chart = Number Defective

$$UCL_{np} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})} \qquad LCL_{np} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})}$$

The c Chart = Number nonconformities with a constant sample size

$\bar{c} = \frac{\text{total nonconformities}}{\text{number of subgroups}}$

$$UCL_c = \bar{c} + 3\sqrt{\bar{c}} \qquad LCL_c = \bar{c} - 3\sqrt{\bar{c}}$$

The u Chart = Number nonconformities with varying sample size

$\bar{u} = \frac{\text{total nonconformities}}{\text{total units inspected}}$

$$UCL_u^* = \bar{u} + 3\sqrt{\frac{\bar{u}}{n}} \qquad LCL_u^* = \bar{u} - 3\sqrt{\frac{\bar{u}}{n}}$$

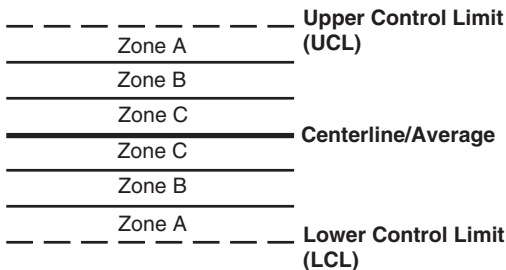
*This formula creates changing control limits. To avoid this, use Average Sample Sizes $\sqrt{\bar{n}}$ for those samples that are $\pm 20\%$ of the average sample size. Calculate individual limits for the samples exceeding $\pm 20\%$.

INTERPRETING CONTROL CHARTS

The process is said to be “out of control” if:

1.) One or more points fall outside of the control limits or:

2.) When you divide the control chart into zones as follows:

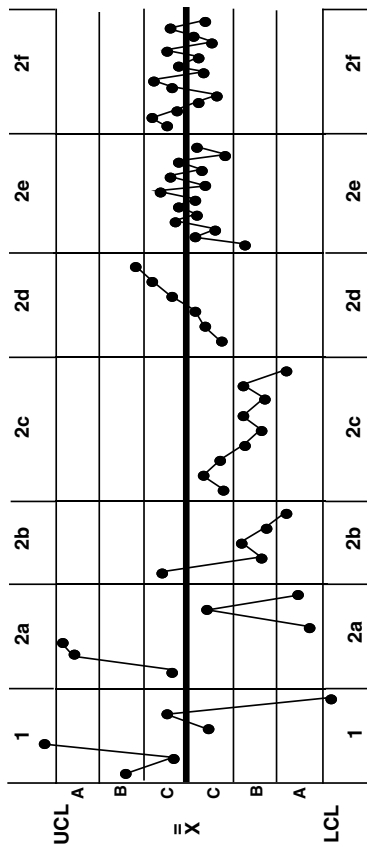


You should take note and examine what has changed and **possibly** make a process adjustment if:

- Two points, out of three successive points, are on the same side of the centerline in Zone A or beyond.
- Four points, out of five successive points, are on the same side of the centerline in Zone B or beyond.
- Nine successive points are on one side of the centerline.
- There are six consecutive points, increasing or decreasing.
- There are fourteen points, in a row alternating up and down.
- There are fifteen points in a row within Zone C (above and below centerline).

(see diagram page 56)

Tests For Control



COMMON QUESTIONS TO ASK WHEN INVESTIGATING AN OUT OF CONTROL PROCESS*

- | | | |
|------------------------------|-----------------------------|---|
| ☐ Yes | ☐ No | Are there differences in the measurement accuracy of instruments used? |
| ☐ Yes | ☐ No | Are there differences in the methods used by different operators? |
| ☐ Yes | ☐ No | Is the process affected by the environment (e.g., temperature, humidity)? |
| ☐ Yes | ☐ No | Has there been a significant change in the environment? |
| ☐ Yes | ☐ No | Is the process affected by tool wear? |
| ☐ Yes | ☐ No | Were any untrained workers involved in the process at the time? |

* **“Out of Control”** as defined by conditions on pages 55 and 56.

COMMON QUESTIONS TO ASK WHEN INVESTIGATING AN OUT OF CONTROL PROCESS*

(Continued)

- | | | |
|------------------------------|-----------------------------|--|
| ☐ Yes | ☐ No | Has there been a change in the source for raw materials? |
| ☐ Yes | ☐ No | Is the process affected by operator fatigue? |
| ☐ Yes | ☐ No | Has there been a change in maintenance procedures? |
| ☐ Yes | ☐ No | Is the machine being adjusted frequently? |
| ☐ Yes | ☐ No | Did the samples come from different machines? Shifts? Operators? |
| ☐ Yes | ☐ No | Are operators afraid to report "bad news"? |

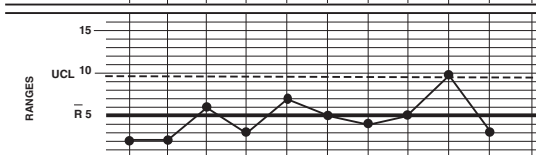
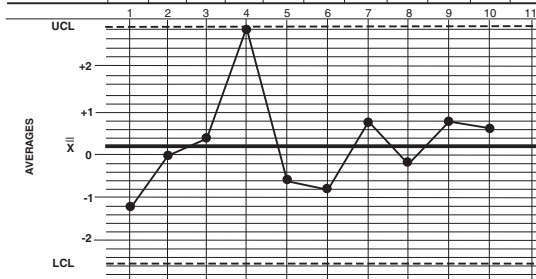
* **"Out of Control"** as defined by conditions on pages 55 and 56.

Control Chart—Mfg. Example

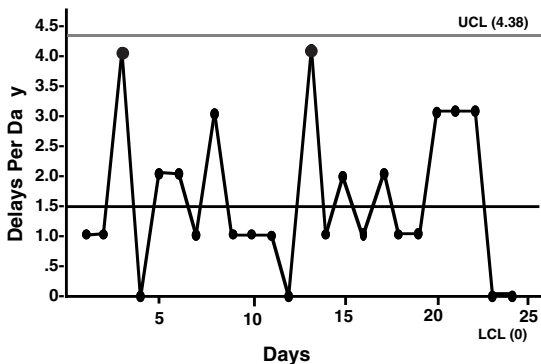
$\bar{X}$ & R Chart



DATE												
TIME		8:00	8:30	9:00	9:30	10:00	10:30	11:00	11:30	12:00	12:30	
SAMPLE MEASUREMENTS	1	-2	+1	+3	+2	-4	0	-1	-3	-6	+2	
	2	-2	0	+1	+3	-3	0	+1	+2	+2	-1	
	3	0	-1	-3	+5	0	-1	+2	-2	0	0	
	4	0	-1	+1	+2	+1	+1	-1	+1	+4	+1	
	5	-2	+1	0	+2	+3	-4	+3	+1	+4	+1	
SUM		-6	0	+2	+14	-3	-4	+4	-1	+4	+3	
AVERAGE, $\bar{X}$		-1.2	0	+4	+2.8	-6	-8	+8	-2	+8	+6	
RANGE, R		2	2	6	3	7	5	4	5	10	3	
NOTES												



Control Chart—Admin./Service Ex. np Chart Operating Room Delays/Day



Control Chart—Daily Example

Commuting Times (min.)—A.M.

STEP 1

Week									
1	2	3	4	5	6	7	8	9	10
55	90	100	70	55	75	120	65	70	100
75	95	75	110	65	85	110	65	85	80
65	60	75	65	95	65	65	90	60	65
80	60	65	60	70	65	85	90	65	60
80	55	65	60	70	65	70	60	75	80
$\bar{X}=71$	72	76	73	71	71	90	74	71	77
$R=25$	40	35	50	40	20	55	30	25	40

STEP 2

$$\bar{\bar{X}} = 74.6$$

$$\bar{R} = 36.0$$

$$n = 5$$

$$k = 10$$

STEP 3

$$\begin{aligned} UCL_{\bar{X}} &= \bar{\bar{X}} + A_2 \bar{R} \\ &= 74.6 + (.58)(36.0) \\ &= 74.6 + 20.88 \\ &= 95.48 \end{aligned}$$

$$\begin{aligned} LCL_{\bar{X}} &= \bar{\bar{X}} - A_2 \bar{R} \\ &= 74.6 - 20.88 \\ &= 53.72 \end{aligned}$$

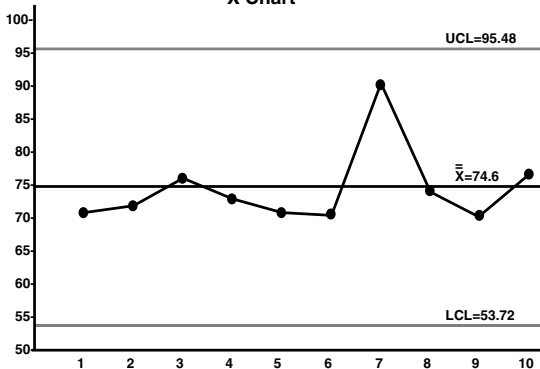
$$\begin{aligned} UCL_R &= D_4 \bar{R} \\ &= (2.11)(36.0) \\ &= 75.96 \end{aligned}$$

$$\begin{aligned} LCL_R &= D_3 \bar{R} \\ &= 0 \end{aligned}$$

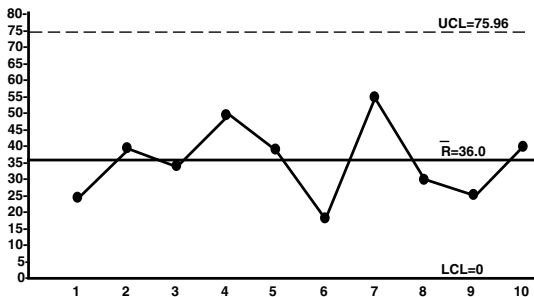
Commuting Times—A.M.

STEP 4

$\bar{X}$ Chart



R Chart





Construction/Interpretation Tips Control Chart

- Generally, collect 20-25 groups of samples before calculating the control limits.
- Upper and Lower Control Limits **MUST** be statistically calculated. Don't confuse them with specification limits, which are based on product requirements.
- Management, through support and action, has the opportunity to control/reduce the natural variation between the control limits.
- Make sure to select the right type of Control Chart for the right type of data (see accompanying chart).
- Data must be kept in the exact sequence as it was gathered, otherwise it is meaningless.
- Do not "tweak" the process beyond standard procedures while you are gathering the data; the data must reflect how it runs *naturally*.



Process Capability: When you need to determine whether the process, given its natural variation, is capable of meeting established (customer) specifications.



PROCESS CAPABILITY

Being in control is not enough. An “in-control” process can produce bad product. True improvement of a process comes from balancing repeatability and consistency with the capability of meeting your customer requirements, otherwise known as Process Capability.

In order to objectively measure the degree to which your process is or is not meeting those requirements, capability indices have been developed to graphically portray that measure. Capability indices let you place the distribution of your process in relation to the specification limits.

FORMULAS FOR CALCULATING PROCESS CAPABILITY INDICES

C_p is a simple process capability index that relates the allowable spread of the specification limits (i.e., the difference between the upper specification limit, USL, and the lower specification limit, LSL) to the measure of the actual, or natural, variation of the process, represented by $6 \hat{\sigma}$, where $\hat{\sigma}$ is the estimated process standard deviation.

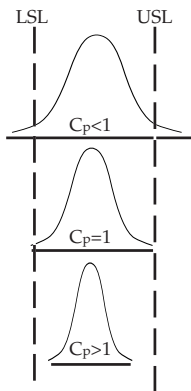
$$C_p = \frac{USL - LSL}{6 \hat{\sigma}}$$

If the process is in statistical control, then $\hat{\sigma}$ can be estimated from the control chart:

$\hat{\sigma} = \bar{R}/d_2$ where $\bar{R}$ = the average of the
subgroup ranges
 d_2 = a tabled value based on the
subgroup sample size

Factors for Estimating $\hat{\sigma}$:

n	d_2	n	d_2
2	1.128	6	2.534
3	1.693	7	2.704
4	2.059	8	2.847
5	2.326	9	2.970
		10	3.078



The process variation exceeds specification => defectives are being made.

The process is just meeting specification. A minimum of .3% defectives will be made, more if the process is not centered.

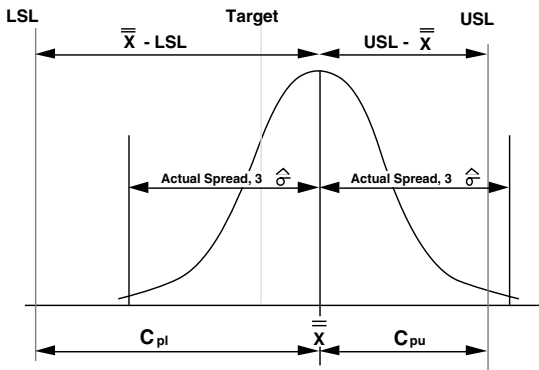
The process variation is less than specification, however, defectives might be made if the process is not centered on the target value.

While C_p relates the spread of the process relative to the specification width, it **DOES NOT** look at how well the process average, $\bar{X}$, is centered to the target value. C_p is often referred to as a measure of process “potential.”

The process capability indices C_{pl} and C_{pu} (for single-

sided specification limits) and C_{pk} (for two-sided specification limits) measure not only the process variation with respect to the allowable specification, they also take into account the location of the process average. C_{pk} is considered a measure of the process “capability” and is taken as the smaller of either C_{pl} or C_{pu} :

$$C_{pl} = \frac{\bar{\bar{X}} - LSL}{3\hat{\sigma}} \quad C_{pu} = \frac{USL - \bar{\bar{X}}}{3\hat{\sigma}} \quad C_{pk} = \min \{ C_{pl}, C_{pu} \}$$



If the process is near normal and in statistical control, C_{pk} can be used to estimate the expected percent of defective material. Estimating the percentages of defective material is beyond the scope of *The Memory Jogger™* and can be found in statistical books.

Process Capability—Mfg. Example

Die Cutting Process



A Control Chart was maintained, producing the following statistics:

$$\begin{array}{ll} \bar{X} = 212.5 & \text{Spec.} = 210 \pm 3 \\ \bar{R} = 1.2 & \text{USL} = 213 \\ n = 5 & \text{LSL} = 207 \end{array}$$

$$\hat{\sigma} = \bar{R} / d_2 = 1.2 / 2.326 = .51$$

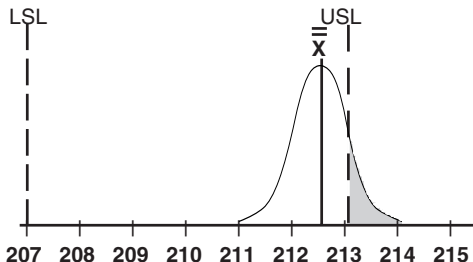
$$C_p = \frac{\text{USL} - \text{LSL}}{6\hat{\sigma}} = \frac{213 - 207}{6(.51)} = \frac{6}{3.06} = 1.96$$

$$C_{pl} = \frac{\bar{X} - \text{LSL}}{3\hat{\sigma}} = \frac{212.5 - 207}{3(.51)} = \frac{5.5}{1.53} = 3.595$$

$$C_{pu} = \frac{\text{USL} - \bar{X}}{3\hat{\sigma}} = \frac{213 - 212.5}{3(.51)} = \frac{0.5}{1.53} = .327$$

$$C_{pk} = \min \{ C_{pl}, C_{pu} \} = .327$$

Since $C_{pk} < 1$, defective material is being made.





Construction/Interpretation Tips Process Capability

- C_p index is limited to two-sided specifications.
- Some companies use the inverse of the C_p ratio

$$C_p = \frac{6\hat{\sigma}}{USL - LSL}$$

- Ask your customer which ratio is being used for interpretation.
- The use of process capability indices assumes realistic and meaningful specifications. Be sure that you and your customer have agreed upon them.
- If $C_{pl} = C_{pu}$, then the process is exactly centered!
- Many companies are establishing specific process capability targets, typically a C_{pk} of 1.33 for supplier qualification, with an expected achievement of a C_{pk} of 2.0 or higher for a long-term target.

OTHER HELPFUL TOOLS

BRAINSTORMING

All of the charting techniques are aids to thinking. They focus the attention of the user on the truly important dimensions of a problem. It is equally important, however, to expand your thinking to include **all** of the dimensions of a problem or solution. Brainstorming is used to help a group create as many ideas in as short a time as possible.

Brainstorming can be used in two ways:

- 1.) **Structured**—In this method, every person in a group must give an idea as their turn arises in the rotation or pass until the next *round*. It often forces even shy people to participate but can also create a certain amount of pressure to contribute.
- 2.) **Unstructured**—In this method, group members simply give ideas as they come to mind. It tends to create a more relaxed atmosphere but also risks domination by the most vocal members.

In both methods the general “rules of the road” are the same. The generally accepted guidelines are as follows:

- Never criticize ideas.
- Write on a flip chart or blackboard **every** idea. Having the words visible to everyone at the same time avoids misunderstandings and reminds others of new ideas.
- Everyone agrees on the question or issue being brainstormed. Write it down, too.
- Record on the flip chart in the words of the speaker; don’t interpret.
- Do it quickly; 5-15 minutes works well.

NOMINAL GROUP TECHNIQUE (NGT)

When selecting which problems to work on and in what order, it often happens that the problem selected is that of the person who speaks the loudest or who has the most authority. This often creates a feeling in the team that *“their”* problem will never be worked on. This can lead to a lack of commitment to work on the problem selected, and the selection of the *“wrong”* problem in the first place. The **Nominal Group Technique** tries to provide a way to give everyone in the group an equal voice in problem selection. The steps in the process are as follows:

1. Have everyone on the team write (or say) the problem that he/she feels is most important. If members of the team do not write the problem out, you need to get them written on a flip chart or blackboard (or somewhere visible), as they are being communicated. If people do produce written problems, collect them when they are finished. Everyone may not feel comfortable writing, but it may make them feel safer talking about sensitive problems at the beginning.
2. Write the problem statements where the team can see them.
3. Check with the team to make sure that the same problem hasn't been written twice (but may be in slightly different words). If a problem is repeated combine them into one item.
4. Ask the team members to write on a piece of paper the letters corresponding to the number of problem statements the team produced. For example, if you ended up with five problem statements, everyone would write the letters “A” through “E” on the paper.
5. Make sure that each problem statement has a letter in front of it. Ask the team members to vote on which problem is most important, then write “5” next to the letter. For example, the problem list may look like this:
 - A. Space
 - B. Safety
 - C. Housekeeping
 - D. Quality going down
 - E. No preventive maintenance

Each member's paper would look like this:

- A. _____
- B. _____
- C. _____
- D. _____
- E. _____

So, if someone thought "*quality is going down*" was the #1 problem, it would look like this:

- A. _____
- B. _____
- C. _____
- D. 5
- E. _____

Everyone then has to complete the list by voting what's second most important, third most important, etc.

- A. 2, 5, 2, 4, 1
- B. 1, 4, 5, 5, 5
- C. 4, 1, 3, 3, 4
- D. 5, 2, 1, 1, 2
- E. 3, 3, 4, 2, 3

An alternative ranking approach involves the "one half plus one" rule. Especially when dealing with many items, it may be necessary to limit the items to be considered. This rule suggests ranking only one half of the items plus one. For example, if 20 items were generated, team members would rate only 11 ideas.

6. Add up each line of numbers across. The item with the **highest** number is the most important one to the total team. In this case "B" (Safety) would be the most important item with a total of "20." You would add up the numbers for each item and put them in order.
7. You would then work on item B first, and then move through the list.

FORCE FIELD ANALYSIS

How does change occur, either personally or organizationally? It's a dynamic process. It suggests movement, either from "*time A*" to "*time B*" or "*condition x*" to "*condition y*." etc. Where does the energy for this "*movement*" come from? One approach is to view change as the result of a struggle between forces that are seeking to upset the status quo. This view is taken in the work of Kurt Lewin, who developed a technique called "*Force Field Analysis*." In it, Lewin proposed that "*driving forces*" move a situation toward change while "*restraining forces*" block that movement. When there is no change, the opposing forces are equal or the restraining forces are too strong to allow movement.

Consider the practical example of "*losing weight*":

Driving Forces	Restraining Forces
Health threat →	← Lack of time
Cultural obsession with being thin →	← Genetic traits
Plenty of thin role models →	← Unsympathetic friends & family
Embarrassment →	← Lack of money for exercise
Negative self-image →	← Lack of interest
Positive attitude toward exercise →	← Bad advice
Lack of temptation →	← Years of bad eating habits
Clothes don't fit →	← Amount of sugar in any prepared food

If the restraining forces are stronger than the driving forces, then the desired change will not happen. It stands to reason that some change (lost weight) will occur if the driving forces are more powerful than those on the restraining side of the ledger.

Why does Force Field Analysis help make change happen?

- 1.) It forces people to think together about all the facets of a desired change; it encourages creative thinking.
- 2.) It encourages people to agree about the relative priority of factors on each side of the "balance sheet." The team can use the Nominal Group Technique to reach consensus quickly.
- 3.) It provides a starting point for action.

How does Force Field Analysis accomplish this last task? One can approach change either from the perspective of strengthening *"driving forces"* or reducing the *"restraining forces."* Strengthening the positive often has the unexpected result of reinforcing the negative. Have you ever seen a situation where someone is told repeatedly that *"X, Y, or Z"* is bad for him or her? Instead of the desired improvement, it often strengthens resistance. It has been shown that the most effective tactic is to diminish or eliminate a restraining force. In our example, it would be much more helpful to deal with *"lack of time"* than to constantly remind someone that his or her *"clothes don't fit"* like they used to.

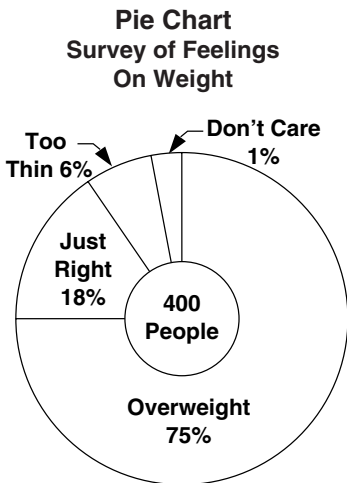
If done honestly, Force Field Analysis can be a helpful aid to thinking and a strategic tool for change.

MANAGEMENT PRESENTATION CHECKLIST

- Decide who the key decision makers are.
- Invite the key decision makers to the presentation.
 - By memo if appropriate
 - By phone the day before to get confirmation
- Arrange for the room.
 - Right size; large enough to be comfortable; not so large that group members are lost
 - Right atmosphere; the ideal is bright, quiet, clean, and informal
- Arrange for audiovisual (AV) equipment.
 - Flip chart
 - Overhead projector
 - Screen
 - Markers
- Decide which charts will help the presentation.
- Prepare charts and AV materials.
 - Assign responsibility
 - Keep AV materials few in number and simple in design
- Assign presentation responsibilities.
- At the presentation:
 - Speak clearly and slowly
 - Listen to questions carefully; you will get an idea of what to stress in the presentation
 - Summarize recommendations simply and clearly

PIE CHART

Pie Charts are simply graphs in which the entire circle represents 100% (not 360 degrees) of the data to be displayed. The circle (pie) is divided into percentage *slices* that clearly show the largest shares of data. This is useful in the same way as a **Pareto Chart**. The Pie Chart is sometimes even more useful since it is widely used to display data on TV or in the newspapers. As with all other graphs, be sure to clearly mark the subject matter, dates if needed, the percentages within the *slices*, and what each slice represents.

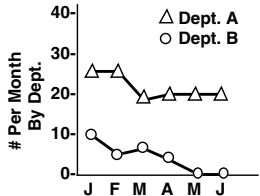
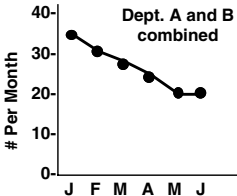


STRATIFICATION*

A technique called Stratification is often very useful in analyzing data to find improvement opportunities. Stratification helps analyze cases in which data actually masks the real facts. This often happens when the recorded data is from many sources but is treated as one number.

For example, data on minor injuries for a plant may be recorded as a single figure either rising or falling. But that number is actually the sum total of injuries:

- By Type: cuts and burns
- By Location: eyes, hands, feet, etc.
- By Department: maintenance, shipping, production, etc.

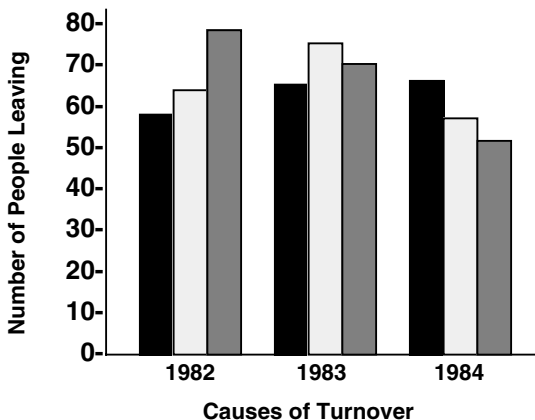


Stratification breaks down single numbers into meaningful categories or classifications to focus corrective action.

*See "Different Uses Of A Pareto Chart," pages 20-22, for stratification in pareto analysis. Also see pages 172-174 of the Western Electric Statistical Quality Control Handbook for examples of stratification in control chart patterns.

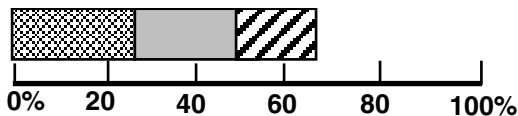
ADDITIONAL BAR GRAPHS

Compound Bar Graph



-  Voluntary Quit
-  Dismissal
-  Lack of Work

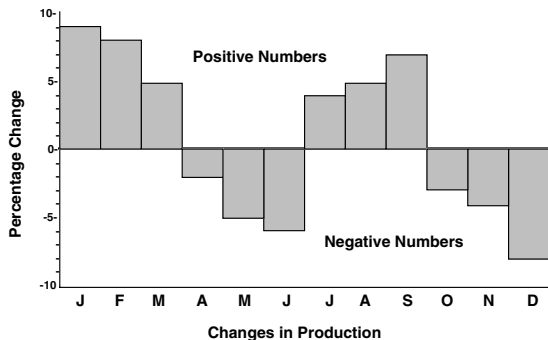
Horizontal Bar Graph



Products As A Percent Of Sales



Negative Numbers On A Bar Graph





Construction/Interpretation Tips Data Points To Ponder

- The aim of data-based problem solving is not to gather **more** data but **meaningful** data.
- The collection and proper use of data short-circuits much of the interpersonal conflict that happens in groups.
- Data can be used to:
 - Understand the current **actual** situation (good and bad)
 - Regulate and modify the process
 - Accept or reject a product or process
- Bad data is worse than no data.
- Data should ideally be based on a truly random sample in which each event or piece has an equal chance of being observed or selected.
- Data that is compared must be gathered **consistently**.
- There are two major types of data, **measured/continuous** and **counted/discrete**.

Measured/Continuous Data that's measured on a continuous scale such as length, weight, time, or temperature.

Counted/Discrete Data that is an accumulation of observations of a particular characteristic such as the number of defects, units sold, or typing errors.

- Every data collection document should include:
 - Name of person(s) collecting data
 - Date of collection
 - Time span covered, e.g., time of day, shift
 - Location of data collection, e.g., department, office
 - Instruments (if applicable) or any other methods used



Construction/Interpretation Tips Common Mistakes To Avoid

- Don't bias the results by the method of sampling. Try to gather sample data in as random a way as possible, e.g., don't take all the pieces from the top of the box.
- Don't confuse control limits with specification limits. Control limits are statistically determined while specification limits are based on what is needed or desired.
- Don't make it more complicated than it needs to be. Use the simplest appropriate tool.
- Don't collect too much or too little data. Don't collect data for a week when a day will do or vice versa.
- Don't overcomplicate graphs. Keep them simple and clear so that the message is apparent to the viewer.
- Don't confuse samples with populations.
- Don't blindly interpret graphs the same way in different situations. Use common sense, e.g., frequency of occurrence may not always be the most significant measure under the Pareto principle. Don't stand on a single statistic. Have other supporting evidence, e.g., find the range, not just the mean.
Don't hesitate to seek help when a situation seems
- too complex or confusing for you to handle. Many companies today have professional statisticians who can help you collect and analyze information in the most efficient and effective way possible.

GLOSSARY OF TERMS USED IN STATISTICAL PROCESS CONTROL

***ATTRIBUTES** are qualitative data that can be counted for recording and analysis. Examples include characteristics such as the presence of a required label and the installation of all required fasteners. Other examples may include characteristics that are inherently measurable (i.e., could be treated as variables). Where the results are recorded in a simple yes/no fashion, (such as acceptability of a shaft diameter when measured on a go/no-go gage), p, np, c, and u charts are used rather than an $\bar{X}$ & R chart.

AVERAGE or mean is the most common expression of the centering of a distribution. It is signified by $\bar{X}$ and is calculated by totaling the observed values and dividing by the number of observations

$$\bar{X} = \frac{(X_1 + X_2 + \dots + X_n)}{n}$$

BIMODAL DISTRIBUTION is one which has two identifiable curves within it, indicating a mixing of two populations such as different shifts, machines, workers, etc.

***COMMON CAUSE** is a source of variation that is always present; part of the random variation inherent in the process itself. Its origin can usually be traced to an element of the system which only management can correct.

***CONTROL CHART** is a graphic representation of a characteristic of a process, showing plotted values of some statistic gathered from that characteristic, and one or two control limits. It has two basic uses: as a judgment to determine if a process was in control, and as an aid in achieving and maintaining statistical control.

***CONTROL LIMIT** is a line (or lines) on a control chart used as a basis for judging the significance of the variation from subgroup to subgroup. Variation beyond a control limit is evidence that special causes are affecting the process. Control limits are calculated from process data and are not to be confused with engineering specifications.

***DETECTION or inspection** is a past-oriented strategy that attempts to identify unacceptable output after it has been produced and separate it from the good output. (See also PREVENTION)

DISTRIBUTION is the population (universe) from which observations are drawn, categorized into cells, and form identifiable patterns. It is based on the concept of variation that states that anything measured repeatedly will arrive at different results. These results will fall into statistically predictable patterns. A *bell-shaped curve* (normal distribution) is an example of a distribution in which the greatest number of observations fall in the center with fewer and fewer observations falling evenly on either side of the average.

FORCE FIELD ANALYSIS is a technique developed by Kurt Lewin that displays the Driving (Positive) and Restraining (Negative) forces surrounding any change. This is displayed in a "balance sheet" format.

FREQUENCY DISTRIBUTION is a statistical table that presents a large volume of data in such a way that the central tendency (average/mean/median) and distribution are clearly displayed.

NOMINAL GROUP TECHNIQUE is a weighted ranking technique that allows a team to prioritize a large number of issues without creating "winners" and "losers."

***NONCONFORMITIES** are specific occurrences of a condition that does not conform to specifications or other inspection standards; sometimes called discrepancies or defects. An individual nonconforming unit can have the potential for more than one nonconformity (e.g., a door could have several dents and dings; a functional check of a carburetor could reveal any of a number of discrepancies). The c and u charts are used to analyze systems producing nonconformities.

POPULATION is the universe of data under investigation from which a sample will be taken.

***PREVENTION** is a future-oriented strategy that improves quality by directing analysis and action toward correcting the production process. Prevention is consistent with a philosophy of never-ending improvement.

***PROCESS** is the combination of people, machine and equipment, raw materials, methods, and environment that produces a given product or service.

PROCESS CAPABILITY is the measured, built-in reproducibility (consistency) of the product turned out by the process. Such a determination is made using statistical methods, not wishful thinking. The statistically determined pattern or distribution can only then be compared to specification limits to decide if a process can consistently deliver product within those parameters.

RANGE is a measure of the variation in a set of data. It is calculated by subtracting the lowest value in the data set from the highest value in that same set.

RUNS are the patterns in a Run Chart or Control Chart within which a number of points line up on only one side of the central line. Beyond a certain number of consecutive points (statistically based) the pattern becomes unnatural and worthy of attention.

***SAMPLE** is one or more individual events or measurements selected from the output of a process for purposes of identifying characteristics and performance of the whole.

***SIGMA** $\hat{\sigma}$ is the Greek letter used to designate the estimated standard deviation.

***SPECIAL CAUSE** is a source of variation that is intermittent, unpredictable, unstable; sometimes called an assignable cause. It is signalled by a point beyond the control limits.

***SPECIFICATION** is the engineering requirement for judging acceptability of a particular characteristic. Chosen with respect to functional or customer requirements for the product, a specification may or may not be consistent with the demonstrated capability of the process (if it is not, out-of-specification parts are certain to be made). A specification should never be confused with a control limit.

***STANDARD DEVIATION** is a measure of the spread of the process output or the spread of a sampling statistic from the process (e.g., of subgroup averages), denoted by the Greek letter $\hat{\sigma}$ (sigma) for the estimated standard deviation.

***STATISTICAL CONTROL** is the condition describing a process from which all special causes have been removed, evidenced on a control chart by the absence of points beyond the control limits and by the absence of non-random patterns or trends within the control limits.

***STATISTICAL PROCESS CONTROL** is the use of statistical techniques such as Control Charts to analyze a process or its output so as to take appropriate actions to achieve and maintain a state of statistical control and to improve the capability of the process.

STRATIFICATION is the process of classifying data into subgroups based on characteristics or categories.

TRENDS are the patterns in a Run Chart or Control Chart that feature the continued rise or fall of a series of points. Like Runs, attention should be paid to such patterns when they exceed a predetermined number (statistically based).

***VARIABLES** are those characteristics of a part which can be measured. Examples are length in millimeters, resistance in ohms, closing effort of a door in kilograms, and the torque of a nut in foot pounds. (See also ATTRIBUTES)

***VARIATION** is the inevitable difference among individual outputs of a process. The sources of variation can be grouped into two major classes: Common Causes and Special Causes.

***From Ford Motor Company's Q101 (1983).** This is a handbook of quality requirements for its manufacturing plants and outside vendors.

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